

Jie Li

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Research Interests

My research focuses on high-performance computing (HPC) and AI infrastructure, with an emphasis on observability, resource management, and energy efficiency. Building on this systems foundation, my research agenda extends to AI agents for HPC operation and management, the security of agents acting within HPC environments, and the integration of HPC with quantum computing.

- **High-performance computing:** system monitoring, workload characterization, scheduling, disaggregated memory, and energy-aware resource management.
- **AI infrastructure:** LLM serving, KV cache and memory management, and performance and power characterization of AI workloads.
- **AI agents for HPC operations:** agent-assisted cluster operation and management, including ongoing development for the NSF REPACSS system.
- **AI agent security in HPC:** evaluating and constraining agent behavior in shared computing environments.
- **HPC and quantum computing:** quantum arithmetic and polynomial synthesis, with broader interests in hybrid quantum-classical workflows and HPC integration.

Education

May 2024	Ph.D., Computer Science , Texas Tech University, Lubbock, TX Dissertation: <i>Optimizing High-Performance Computing Systems: Insights from System Monitoring, Workload Management, and Scheduling Strategies</i> . Advisor: Prof. Yong Chen.
August 2019	M.S., Computer Science , Texas Tech University, Lubbock, TX Thesis: <i>PIMS: A Lightweight Processing-in-Memory Accelerator for Stencil Computations</i> . Advisor: Prof. Yong Chen.
May 2012	B.A., Architecture , Huaqiao University, Xiamen, China

Academic Appointments and Research Experience

Jul 2026 – present	Research Assistant Professor , Department of Computer Science, Texas Tech University • Lead research in HPC and AI infrastructure, spanning observability, resource and data management, energy-aware computing, cyberinfrastructure security, and AI-driven system management.
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- Initiated a research direction on AI agents for autonomous HPC operation and management, using REPACSS as a production testbed.
- Contribute to the operation, optimization, and research use of the NSF REPACSS cluster (\$12.25M), including system software integration, performance analysis, and user support.
- Supervise graduate and undergraduate researchers and develop proposals on secure, autonomous, and energy-efficient computing infrastructure.

Jan 2026 – present

Assistant Director, NSF Cloud and Autonomic Computing Center (CAC IUCRC), Texas Tech site

- Coordinate faculty, student teams, industry members, and external collaborators in developing and reviewing research projects for this NSF Industry–University Cooperative Research Center.
- Help organize semiannual Industry Advisory Board meetings, including project planning, presentation preparation, and follow-up with industry participants.
- Mentor students preparing presentations, posters, and reports for industry review; support proposal development, center reporting, and cross-institutional coordination.

Jun 2024 – Jun 2026

Postdoctoral Researcher, Texas Tech University

- Co-designed and built the NSF REPACSS cluster (\$12.25M) from the ground up, contributing to system architecture, deployment, operation, and performance optimization.
- Served as technical lead of the Data-Intensive Scalable Computing Laboratory, coordinating research activities, student projects, and collaborative system development.
- Co-PI on the NSF CICI proposal SHIELD (2025, not funded), which proposed layered defense mechanisms for open-science cyberinfrastructure.

Sep 2019 – May 2024

Research Assistant, Texas Tech University

- Across graduate study (2019–2024), published 15 peer-reviewed papers (7 first-author) in venues including CLUSTER, ISC, ICPP, MemSys, IEEE CLOUD, IEEE BigData, and IEEE Transactions on Computers.
- Developed and maintained MonSTer (later adopted by Dell’s Omnia project), the Disaggregation-Aware Scheduler, and xBGAS simulation tools, supporting research in HPC monitoring, workload characterization, scheduling, and memory systems.
- Mentored students on HPC monitoring, data management, and workload analysis, including a master’s thesis that led to a CLOUD’23 publication.

Summers 2021, 2022,
2023

Graduate Student Intern, Lawrence Berkeley National Laboratory, Berkeley, CA
Mentors Brandon Cook and Georgios Michelogiannakis, in John Shalf’s group.

- Designed and implemented pipelines integrating LDMS, DCGM, and Slurm telemetry from NERSC's Cori and Perlmutter supercomputers.
- Applied machine learning and deep learning to classify, characterize, and predict HPC job behavior from large-scale time-series telemetry.
- Led development of the open-source Disaggregation-Aware Scheduler; internship research resulted in first-author publications at ISC'23 and CLUSTER'24.

Grants and Proposals

2023	REPACSS: Empowering Scientific Discovery through Renewable Energy Powered Advanced Computing Systems and Services, National Science Foundation, Category II · Funded · Total award \$12,250,000 Role: Contributor to proposal development and infrastructure implementation (PI: Yong Chen; Co-PI: Alan Sill). Contributed proposal sections on data center monitoring and remote control.
2025	SHIELD: Strengthening High-Performance Infrastructure with Enhanced Layered Defense, NSF CICI: UCSS · Not funded · Requested \$600,000 Role: Co-PI (PI: Yong Chen). Contributed project vision and technical approach for HPC cybersecurity.
2020	DA Vinci: An Integrated Data Collection, Automation, and Visualization Framework for HPC Systems, NSF Frameworks · Not funded · Requested \$1,000,000 Role: Lead contributor (PI: Yong Chen; Co-PI: Alan Sill). Led core framework design and research methodology.

Peer-Reviewed Publications

TokenPowerBench: Benchmarking the Power Consumption of LLM Inference
 Chenxu Niu, Wei Zhang, Jie Li, Yongjian Zhao, Tongyang Wang, Xi Wang, and Yong Chen
In Proceedings of the AAAI Conference on Artificial Intelligence (AAAI'26), 2026
 (Acceptance Rate: 4,167/23,680=17.6%)

Power-Centric Observability for HPC Systems: Design, Deployment, and Evaluation on REPACSS
 Yongjian Zhao, Jie Li, Chenxu Niu, Alan Sill, and Yong Chen
In Practice and Experience in Advanced Research Computing (PEARC'26), 2026

OCEAN: Open-Source CXL Emulation for Hyperscale Architecture and Networking

Yiwei Yang, Mujahid Al Rafi, Jesun Sahariar Firoz, Zhen Peng, Xi Wang, Kenza Bouasker, Sayan Ghosh, [Jie Li](#), Daniel Wang, Dong Li, Hyeran Jeon, Kevin Barker, Nathan Tallent, and Luanzheng Guo
In The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'26), 2026
Accepted

Towards Cycle-accurate Simulation of xBGAS

[Jie Li](#), John D Leidel, Brian Page, and Yong Chen
In 2024 International Conference on Computing, Networking and Communications (ICNC'24), 2024

Scheduling and Allocation of Disaggregated Memory Resources in HPC Systems

[Jie Li](#), George Michelogiannakis, Brandon Cook, John Shalf, and Yong Chen
In 2024 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW'24), 2024

Job Scheduling in High Performance Computing Systems with Disaggregated Memory Resources

[Jie Li](#), George Michelogiannakis, Samuel Maloney, Brandon Cook, Estela Suarez, John Shalf, and Yong Chen
In 2024 IEEE International Conference on Cluster Computing (CLUSTER'24), 2024

Analyzing Resource Utilization in an HPC System: A Case Study of NERSC's Perlmutter

[Jie Li](#), George Michelogiannakis, Brandon Cook, Dulanya Cooray, and Yong Chen
In International Conference on High Performance Computing (ISC'23), 2023
(Acceptance Rate: 21/78=26.9%)

Workload Failure Prediction for Data Centers

[Jie Li](#), Rui Wang, Ghazanfar Ali, Tommy Dang, Alan Sill, and Yong Chen
In 2023 IEEE 16th International Conference on Cloud Computing (CLOUD'23), 2023

Effective Management of Time Series Data

Cristiano E. Caon, [Jie Li](#), and Yong Chen
In 2023 IEEE 16th International Conference on Cloud Computing (CLOUD'23), 2023

Spiro: Order-Preserving Visualization in High Performance Computing Monitoring

Tommy Dang, Ngan VT Nguyen, [Jie Li](#), Alan Sill, and Yong Chen
In International Symposium on Visual Computing (ISVC'23), 2023

JobViewer: Graph-based Visualization for Monitoring High-Performance Computing System

Tommy Dang, Ngan VT Nguyen, [Jie Li](#), Alan Sill, Jon Hass, and Yong Chen
In 2022 IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT'22), 2022

HAM: Hotspot-Aware Manager for Improving Communications With 3D-Stacked Memory

Xi Wang, Antonino Tumeo, John D Leidel, [Jie Li](#), and Yong Chen
IEEE Transactions on Computers (IEEE Trans Comput), 2021

The Gap between Visualization Research and Visualization Software in High-Performance Computing Center

Tommy Dang, Ngan Nguyen, Jon Hass, [Jie Li](#), Yong Chen, and Alan Sill

The Gap between Visualization Research and Visualization Software (VisGap'21), 2021

RadarViewer: Visualizing the Dynamics of Multivariate Data

Ngan Nguyen, Jon Hass, Yong Chen, [Jie Li](#), Alan Sill, and Tommy Dang

In Practice and Experience in Advanced Research Computing (PEARC'20), 2020

MonSTER: An Out-of-the-Box Monitoring Tool for High Performance Computing Systems

[Jie Li](#), Ghazanfar Ali, Ngan Nguyen, Jon Hass, Alan Sill, Tommy Dang, and Yong Chen

In 2020 IEEE International Conference on Cluster Computing (CLUSTER'20), 2020

(Acceptance Rate: 27/132=20.5%)

MAC: Memory Access Coalescer for 3D-Stacked Memory

Xi Wang, Antonino Tumeo, John D Leidel, [Jie Li](#), and Yong Chen

In Proceedings of the 48th International Conference on Parallel Processing (ICPP'19), 2019

(Acceptance Rate: 106/405=26.2%)

PIMS: A Lightweight Processing-in-Memory Accelerator for Stencil Computations

[Jie Li](#), Xi Wang, Antonino Tumeo, Brody Williams, John D Leidel, and Yong Chen

In Proceedings of the International Symposium on Memory Systems (MemSys'19), 2019

MTSAD: Multivariate Time Series Abnormality Detection and Visualization

Vung Pham, Ngan Nguyen, [Jie Li](#), Jon Hass, Yong Chen, and Tommy Dang

In 2019 IEEE International Conference on Big Data (BigData'19), 2019

Preprints

From Tensor Buffer to Distributed Memory Hierarchy: A Survey of KV Cache Management for LLM Serving

[Jie Li](#), Tongyang Wang, and Yong Chen

arXiv preprint arXiv:2607.02574, 2026

Trusted Credentials, Untrusted Behavior: Benchmarking LLM-Agent Security in High-Performance Computing

[Jie Li](#)

arXiv preprint arXiv:2607.18485, 2026

ARcode: HPC Application Recognition Through Image-encoded Monitoring Data

[Jie Li](#), Brandon Cook, and Yong Chen

arXiv preprint arXiv:2301.08612, 2023

Manuscripts Under Submission or Revision

- Ziqing Guo, **Jie Li**, Ziwen Pan, and Yong Chen. NNQA: Neural-Native Quantum Arithmetic for End-to-End Polynomial Synthesis. Submitted to AAAI-27.
- Tongyang Wang, **Jie Li**, Chenxu Niu, and Yong Chen. ECHO: Online Control of CPU Frequency for Energy-Efficient High-Performance Computing.
- Batuhan Sencer, Chenxu Niu, **Jie Li**, and Yong Chen. Slimming Models, Saving Watts: Understanding and Modeling the Impact of Knowledge Distillation on GPU Clusters.

Open-Source Research Software

MonSTer

Out-of-the-box HPC monitoring framework. Published at CLUSTER'20; adopted by Dell's Omnia project.

Disaggregation-Aware Scheduler

Simulation framework for job scheduling in HPC systems with disaggregated memory. Used in CLUSTER'24 and IPDPSW'24 publications.

xBGAS REV-CPU

Cycle-accurate xBGAS simulation extension based on REV-CPU. Developed with Tactical Computing Laboratories; published at ICNC'24.

xBGAS Runtime

Lightweight runtime supporting global address space extensions for RISC-V-based xBGAS simulation. Developed with Tactical Computing Laboratories.

Teaching

Operating systems, parallel and high-performance computing, computer architecture, distributed and cloud computing, and systems for machine learning at the undergraduate and graduate levels. I am also interested in developing new graduate seminars on **AI infrastructure** and on **AI agents for computing systems**.

Fall 2025

Invited Lecturer and Programming Project Designer, *Parallel Processing* (graduate), Texas Tech University

Delivered four invited lectures and designed programming projects for 27 students; topics included job scheduling, compilation and job submission, and OpenMP.

Research Mentoring

Graduate Students

- Summer 2026 – present **Jaechang Kim**, M.S. student
AI agents for REPACSS cluster operation and management, automating Warewulf, Ansible, Spack, and Slurm workflows.
- Fall 2025 – present **Rupak Kadel**, Ph.D. student
Improving HPC monitoring for anomaly and intrusion detection.
- Fall 2022 **Cristiano Caon**, M.S.
Data volume reduction and query optimization in time-series databases; independent study (CS 7000) and master's thesis.
Outcome: CLOUD'23 paper
- Spring 2021 **Aniruddh Sanjaysinh Chavda and Huyen Nguyen**, M.S. and Ph.D.
Usage behavior analysis by clustering job accounting data; Advanced Operating Systems course project.
- Spring 2021 **Ruonan Wu**, M.S.
Job accounting data analysis for the QuanaH cluster; Advanced Operating Systems course project.
- Spring 2020 **Ashhrita Puradamane Balachandra**, M.S.
Improving InfluxDB query performance; Advanced Operating Systems course project.

Undergraduate Students

- Spring 2022 **Yusheng Han and Zachary Kay**
Running HPC applications and analyzing performance on the RedRaider cluster; independent study (CS 4000).
- Spring 2021 **Casey Root**
Monitoring queue status through the Slurm REST API; independent study (CS 4000).

Presentations

- 2021 – 2024 **Integrated Data Collection and Visualization Framework for Data Centers based on Telemetry Model**, NSF CAC Industry Advisory Board Conferences
J. Li et al. Four presentations: Lubbock, TX (May 2021); Denton, TX (Nov 2022); Tucson, AZ (May 2023); Lubbock, TX (Dec 2024).

Apr 2024	Towards Cycle-Accurate Simulation of xBGAS , Latch-Up 2024, Cambridge, MA J. Li and Y. Chen.
Jul 2023	Workload Failure Prediction for Data Centers , IEEE CLOUD'23, Chicago, IL J. Li, R. Wang, G. Ali, T. Dang, A. Sill, and Y. Chen.
Nov 2021	Advanced Visualization and Data Analysis of HPC Cluster and User Application Behavior , ACM/IEEE SC'21 J. Li, N. Nguyen, G. Ali, T. Dang, Y. Chen, and A. Sill.
2020	MonSTer: An Out-of-the-Box Monitoring Tool for High Performance Computing Systems , IEEE CLUSTER'20 J. Li, G. Ali, N. Nguyen, J. Hass, A. Sill, T. Dang, and Y. Chen.
Oct 2019	PIMS: A Lightweight Processing-in-Memory Accelerator for Stencil Computations , MemSys'19 J. Li, X. Wang, A. Tumeo, B. Williams, J. D. Leidel, and Y. Chen.

Refereed Posters

2022	A Holistic View of Memory Utilization on Perlmutter , SC'22 J. Li, B. Cook, G. Michelogiannakis, and Y. Chen.
2021	Detecting and Identifying Applications by Job Signatures , SC'21 J. Li, B. Cook, and Y. Chen.
2019	Memory Hotspot Optimizations for 3D-Stacked Memory , PACT'19 X. Wang, J. Li, A. Tumeo, J. D. Leidel, and Y. Chen.

Awards and Honors

2022	Best Poster Award NSF Cloud and Autonomic Computing Center Industry Advisory Board Conference
Summer 2019	Summer Thesis/Dissertation Research Award (\$2,300) Texas Tech University Graduate School

Professional Service

Program committees	NeurIPS 2026 · AAAI 2026
Conference reviewer	IEEE ISCAS 2026 · IEEE BigData 2022, 2023, 2025 · IEEE/ACM CCGrid 2024

Journal reviewer	IEEE Computer Architecture Letters (2025) · The Journal of Supercomputing (2023)
Sub-reviewer	IEEE IPDPS 2023 · IEEE ICDCS 2022 · ACM/IEEE SC 2022 · International Parallel Data Systems Workshop (PDSW) 2022 · IEEE Smart Data Services 2020
Student volunteer	ACM/IEEE SC'21, St. Louis, MO · ACM/IEEE SC'19, Denver, CO